

The Qang (qg): A Unified Angular-Probability Unit and Metric for Parametric Quantum Circuit Design

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Abstract

In contemporary quantum information processing, the parameterization of single-qubit states relies on polar coordinates on the Bloch sphere, while measurement outcomes are expressed as statistical expectation values or collapse probabilities. We propose the qang (symbol: qg), a normalized dimensionless unit designed to bridge the geometric rotation angle of a state vector directly with its measurement probability distribution. We present two formal operational definitions: the **Polar Bias Qang** (qg_Z), anchored to the expectation value of the Pauli-Z observable, and the **Superposition Dispersion Qang** (qg_S), mapping polar deviation to the binary Shannon entropy of measurement outcomes. We demonstrate its utility in gradient-based optimization algorithms (VQE, QAOA) and discuss how defining a domain-specific unit streamlines compilation, noise characterization, and hardware benchmarking.

1. Introduction and Motivation

In quantum computing and quantum state engineering, parameterized single-qubit rotations serve as the fundamental building blocks of variational circuits (Ansätze), quantum machine learning models, and fault-tolerant compilation pipelines.

Under the standard Bloch sphere representation, an arbitrary pure single-qubit state is parameterized as:

$$|\psi\rangle = \cos(\theta / 2)|0\rangle + e^{i\varphi} \sin(\theta / 2)|1\rangle$$

where $\theta \in [0, \pi]$ represents the polar angle and $\varphi \in [0, 2\pi)$ is the azimuthal phase angle.

When executing algorithms, researchers repeatedly oscillate between two distinct mathematical representations:

1. Continuous physical control parameters (microwave pulse duration, phase shifts, rotation angles in radians or degrees).
2. Probabilistic projection profiles obtained through Born's rule upon measurement:
 - $P(|0\rangle) = |\langle 0|\psi\rangle|^2 = \cos^2(\theta / 2)$
 - $P(|1\rangle) = |\langle 1|\psi\rangle|^2 = \sin^2(\theta / 2)$

Currently, bridging these domains requires explicit trigonometric conversions. To provide an intuitive, standardized metric that encapsulates both geometric displacement and probabilistic outcome bias, we introduce the **qang** (abbreviated as **qg**).

2. Mathematical Formalization

We formulate the qang under two complementary definitions:

2.1 Polar Bias Qang (qg_Z)

The primary formulation of the qang quantifies the measurement bias between computational basis states $|0\rangle$ and $|1\rangle$ as a function of the polar angle θ :

$$\text{qg_Z}(\theta) = P(|0\rangle) - P(|1\rangle) = \cos^2(\theta / 2) - \sin^2(\theta / 2) = \cos(\theta)$$

Equivalently, in terms of quantum mechanics operators, qg_Z corresponds to the expectation value of the Pauli-Z operator:

$$\text{qg_Z} = \langle \psi | \sigma_z | \psi \rangle$$

Inversion Relation

Given a target polarization or expectation value $\text{qg_Z} \in [-1, +1]$, the required polar rotation angle θ is directly recovered via:

$$\theta = \arccos(\text{qg_Z})$$

Standard Anchor Points

State Description	Polar Angle (θ)	Qang Value (qg_Z)
North Pole ($ 0\rangle$)	0 rad	+1.0 qg
Equator / Maximal Superposition ($ +\rangle, -\rangle$)	$\pi/2$ rad	0.0 qg
South Pole ($ 1\rangle$)	π rad	-1.0 qg
Canonical T-gate state	$\pi/4$ rad	$\approx +0.7071$ qg

2.2 Measurement-Outcome Entropy Qang (qg_S)

For tasks quantifying quantum coherence loss or state purity relative to maximal superposition, the entropic formulation normalizes the state's polar projection against the binary Shannon entropy of the Z-basis measurement outcome distribution. Note that this is the classical Shannon entropy of the Born-rule outcome probabilities, not the von Neumann entropy of the quantum state itself — for any pure state $|\psi\rangle$, $S(\rho) = -\text{Tr}(\rho \log \rho)$ is identically zero regardless of θ , since $\rho = |\psi\rangle\langle\psi|$ has eigenvalues $\{1, 0\}$. The quantity below is the entropy of the classical distribution obtained after measurement:

$$qg_S(\theta) = -P(|0\rangle) \log_2(P(|0\rangle)) - P(|1\rangle) \log_2(P(|1\rangle))$$

In this scale:

- Deterministic classical poles ($|0\rangle, |1\rangle$) correspond to $qg_S = 0.0$ qg.
- Maximum quantum uncertainty ($\theta = \pi/2$) corresponds to $qg_S = 1.0$ qg.

Domain and Invertibility: Because the binary entropy function is symmetric, $H(p) = H(1-p)$, $qg_S(\theta) = qg_S(\pi-\theta)$: a single qg_S value does not uniquely determine θ over the full domain $[0, \pi]$. A well-defined inversion $\theta = H^{-1}(qg_S)$ therefore requires restricting to a half-domain, e.g. $\theta \in [0, \pi/2]$.

State Description	Polar Angle (θ)	Qang Value (qg_S)
North Pole ($ 0\rangle$)	0 rad	0.0 qg

State Description	Polar Angle (θ)	Qang Value (qg_S)
Equator / Maximal Superposition ($ +\rangle, -\rangle$)	$\pi/2$ rad	1.0 qg
South Pole ($ 1\rangle$)	π rad	0.0 qg
Canonical T-gate state	$\pi/4$ rad	≈ 0.601 qg

Table 2. Anchor values of the entropic qang (qg_S). Note that both poles map to the same value (0.0 qg), reflecting the non-invertibility discussed above.

2.3 Relation to Existing Formalisms

It is important to state plainly what is and is not new here. $qg_Z(\theta) = \cos(\theta)$ is numerically identical to $\langle \sigma_z \rangle$, the expectation value of the Pauli-Z operator — equivalently, the z-component of the Bloch vector — which is already a standard quantity throughout quantum information theory (Nielsen & Chuang, 2010). Likewise, $qg_S(\theta)$ is the ordinary binary Shannon entropy $H(p)$ applied to the Born-rule outcome distribution. Neither qg_Z nor qg_S is a new physical quantity.

The contribution proposed in this work is instead a named, normalized engineering unit built on top of these existing quantities — fixed anchor points, an explicit inversion relation, and a millesimal subunit (the milliqang) intended for direct use in circuit design, calibration and optimization tooling. This is analogous to how the decibel (dB) is not a new physical quantity but a scaled, named convention over a pre-existing power ratio, or how the musical "cent" is a named convention over a logarithmic frequency ratio. The qang's value is proposed to lie in this same category: a practical, standardized unit of convenience rather than a physical discovery.

2.4 The qang Gate: Euler-Matrix Realization

Every single-qubit unitary (up to global phase) decomposes, via Euler's formula applied to the matrix exponentials of the Pauli generators, into the standard three-angle Euler rotation sequence $R_z(\varphi) \cdot R_y(\theta) \cdot R_z(\lambda)$:

$$Ry(\theta) = \exp(-i\theta/2 \cdot \sigma_y) = I \cdot \cos(\theta/2) - i \cdot \sigma_y \cdot \sin(\theta/2), \quad Rz(\lambda) = \exp(-i\lambda/2 \cdot \sigma_z) = \text{diag}(\cos(\lambda/2) - i \cdot \sin(\lambda/2), \cos(\lambda/2) + i \cdot \sin(\lambda/2))$$

Substituting the half-angle identities already used for Table 1 ($\cos(\theta/2) = \sqrt{(1+qg_Z)/2}$, $\sin(\theta/2) = \sqrt{(1-qg_Z)/2}$) gives a single closed matrix, the qang gate:

$$U3_qang(qg_Z, \phi, \lambda) = \begin{bmatrix} \sqrt{(1+qg_Z)/2} & -e^{i\lambda} \cdot \sqrt{(1-qg_Z)/2} \\ e^{i\phi} \cdot \sqrt{(1-qg_Z)/2} & e^{i(\phi+\lambda)} \cdot \sqrt{(1+qg_Z)/2} \end{bmatrix}$$

verified against the standard $U3(\theta, \phi, \lambda)$ definition to machine precision (max error $\approx 5 \times 10^{-15}$ over 200 random parameter triples), and reproducing every named single-qubit gate as an anchor: $U3_qang(1, 0, 0) = I$, $U3_qang(0, 0, \pi) = \text{Hadamard}$, $U3_qang(-1, 0, \pi) = \text{Pauli-X}$. This is not a new gate — it is the existing $U3$ gate, re-parameterized so its polar argument reads directly in the measurement-outcome unit qg_Z instead of a raw angle, exactly the same honest caveat as Section 2.3. It generalizes, rather than replaces, the qang repository's existing `FullRQangGate`, which already implements the (qg_Z, ϕ) two-parameter case by decomposing to Qiskit's `UGate`($\theta, \phi, 0.0$) with λ fixed at zero; $U3_qang$ frees λ as a third independent parameter, completing the gate family `FullRQangGate` started.

The same Euler-exponential mechanism extends to the qudit generalization already sketched in Future Research Direction #4: because the $\mathfrak{su}(d)$ Cartan generators h_k commute pairwise, $\exp(-i \sum \gamma_k \cdot h_k)$ factorizes exactly into $\prod \exp(-i \gamma_k \cdot h_k)$, each factor given by Euler's formula termwise (verified: max error $\sim 10^{-16}$ for the $\mathfrak{su}(3)$ case against direct joint matrix exponentiation). The generalized values $qg_Z, k := \langle h_k \rangle$ parameterize this factored form directly, the same way qg_Z parameterizes Ry above.

3. Key Properties and Advantages

1. **Direct Operational Meaning:** Expressing changes in qg immediately informs the circuit designer of the resultant probability shift without requiring mental or algorithmic trigonometric transformation.
2. **Phase Decoupling:** The polar qang cleanly isolates population probability dynamics from phase shifts ϕ , simplifying the tuning of non-commuting operations.
3. **Scale Invariance:** The metric is normalized to $[-1, +1]$, aligning naturally with hardware cost functions, loss landscapes, and neural network activation profiles.

4. Application Domains

4.1 Variational Quantum Eigensolvers (VQE) & QAOA

In parameterized quantum circuits, optimization trajectories often suffer from barren plateaus or ill-conditioned gradients when parameterized purely in radians. Formulating gradients with respect to qang space:

$$\partial E / \partial(\text{qg}) = (\partial E / \partial \theta) * (d\theta / d(\text{qg})) = -(1 / \sin(\theta)) * (\partial E / \partial \theta)$$

provides an alternative parameter landscape that highlights population transfer rates.

Limitation: This reparametrization is not free of cost: since $d(\text{qg_Z})/d\theta = -\sin(\theta)$, the Jacobian vanishes exactly at the poles $\theta = 0, \pi$ — precisely the ± 1.0 qg anchor points in Table 1 — so $\partial E / \partial(\text{qg})$ diverges there. The qang reparametrization trades the known difficulties of θ -space (e.g. barren plateaus) for a coordinate singularity of its own near the poles, and should be used away from that region, or combined with an explicit regularization near $|\text{qg_Z}| \rightarrow 1$.

4.2 Quantum Error Characterization & Calibration

Drift in microwave pulse amplitudes leads to over-rotation or under-rotation errors (e.g., coherent systematic errors). Expressing drift in milliqangs (m-qg) establishes a direct error tolerance threshold on observable fidelity.

Limitation: This tolerance is not uniform across the Bloch sphere: because $|d(\text{qg_Z})/d\theta| = \sin(\theta)$, the qang scale is most sensitive to angular drift at the equator ($\theta = \pi/2$) and least sensitive near the poles, where it approaches zero. A fixed milliqang tolerance therefore corresponds to a much looser angular tolerance near $|0\rangle$ and $|1\rangle$ than near a superposition state, and calibration budgets expressed in milliqangs should state the operating point θ at which they apply.

4.3 Grover's Algorithm and Amplitude Amplification

Grover's algorithm and amplitude amplification more generally stay, at every step, inside a real two-dimensional subspace $\text{span}\{|\alpha\rangle, |\beta\rangle\}$, where $|\alpha\rangle$ is the uniform superposition over the $N-M$ unmarked items and $|\beta\rangle$ is the uniform superposition over the M marked items. This is exactly

the object `qg_Z` was built to describe: $|\psi(\theta)\rangle = \cos(\theta/2)|\alpha\rangle + \sin(\theta/2)|\beta\rangle$, real amplitudes, a single polar angle. Defining θ_0 via $\sin(\theta_0/2) = \sqrt{M/N}$ (the standard Grover angle, written here in the paper's own convention), the state after k applications of the Grover operator sits at $\theta_k = (2k+1)\cdot\theta_0$, so:

$$\text{qg_Z}(\theta_k) = \cos(\theta_k) = 1 - 2 \cdot \text{P_marked}(k)$$

reproduces the entire success-probability trajectory of Grover's algorithm as `qg_Z` evaluated along a sequence of angles advancing by $2\theta_0$ per iteration, verified to machine precision against exact statevector simulation (companion notebook, Section 14). The Section 4.1 gradient result, $|d(\text{qg_Z})/d\theta| = \sin(\theta)$, reappears here: propagating an error in θ_0 (i.e. an inaccurate estimate of M/N) through $(2k+1)$ iterations and then through that same gradient gives the calibration sensitivity of Grover's success probability to a wrong guess of M . This complements, rather than duplicates, the existing `qang.transformations.grover_diffuser_qang` helper, which already expresses the diffuser operator itself as a Householder reflection in `qg`-space; the result here is the complementary iteration-by-iteration trajectory.

Limitation: This reformulation is exact for $M \ll N$, the regime Grover's algorithm targets. The closed-form optimal iteration count $k_{\text{opt}} = \text{round}(\pi/(2\theta_0) - 1/2)$ is a continuous approximation to an integer arg-max; when M/N is not small, θ_0 is large, $\text{P_marked}(k)$ completes multiple oscillation periods within a handful of iterations, and the rounded formula can alias to a local rather than the global maximum (verified numerically: for $N=16$, $M=5$, the formula gives $k_{\text{opt}}=1$ while the true optimum is $k=6$). This is not a flaw introduced by the `qg_Z` reformulation — it is the same well-known coarse-angle behavior of Grover's algorithm itself, now simply visible as a statement about $\text{qg_Z}(\theta_k)$. Estimation work targeting a near-balanced search (M comparable to N) should locate k_{opt} by direct search over the `qg_Z` trajectory rather than the closed-form approximation.

Gate-level realization: This trajectory is not only an abstract statement about `qg_Z` — it is directly buildable as a circuit. The existing `grover_diffuser_qang` helper already expresses the diffuser D as a Householder reflection parameterized in `qg`-space; the oracle O and diffuser D at each iteration can equally be assembled from the `qang` gate of Section 2.4 (`U3_qang`), giving a circuit-level realization of the $\text{qg_Z}(\theta_k)$ trajectory above rather than only a statevector-level one.

4.4 T1 Relaxation and Randomized Benchmarking

Because $qg_Z := \langle \sigma_z \rangle$ is well-defined for any density matrix, not only for pure Bloch-sphere states (Future Research Direction #4, extension to mixed states), it extends without modification to two of the standard calibration experiments already named in Section 4.2. For a qubit prepared in $|1\rangle$ and left to relax, population decays as $p_1(t) = p_1(0) \cdot \exp(-t/T_1)$; defining $qg_Z_eff(t) := 1 - 2 \cdot p_1(t)$ as this same unit applied at each instant of time rather than to a single rotation angle gives:

$$1 - qg_Z_eff(t) = 2 \cdot p_1(0) \cdot \exp(-t/T_1) \Leftrightarrow \ln(1 - qg_Z_eff(t)) = \ln(2 \cdot p_1(0)) - t/T_1$$

which is linear in t , turning the usual nonlinear exponential fit for T_1 into an ordinary linear regression on log-distance-from-the-pole. The idealized (SPAM-free) single-qubit randomized-benchmarking survival probability after m random Cliffords, $F(m) = 0.5 + 0.5 \cdot p^m$ (Knill et al., 2008), collapses the same way: defining $qg_Z_RB(m) := 2 \cdot F(m) - 1$ gives exactly $qg_Z_RB(m) = p^m$, so $\ln(qg_Z_RB(m)) = m \cdot \ln(p)$ is linear in the sequence length m , and the average per-Clifford gate infidelity $r = (1-p)/2$ follows directly. Both reformulations were verified against exact (noiseless) algebra and against finite-shot synthetic data (companion notebook, Section 15).

Limitation: Under realistic finite-shot noise, the closed-form qg_Z -linearized estimator is measurably less statistically efficient than a properly weighted nonlinear fit with the same number of free parameters — log-transforming binomial-noisy data introduces heteroscedastic error that an ordinary least-squares fit does not account for (roughly double the standard deviation of the extracted decay constant, in both the T_1 and RB cases tested). What it offers instead is a single closed-form linear solve with no initial guess and no nonlinear-solver convergence risk, which is a genuine practical advantage at the scale of calibrating many qubits automatically, but a statistical-efficiency trade-off rather than a strict improvement.

4.5 Scope Boundary: Phase-Estimation Algorithms (Shor, Quantum Counting)

Not every quantum algorithm admits a qg_Z reformulation, and Shor's algorithm is a useful negative case to state explicitly rather than leave as an unexamined gap. Shor's algorithm extracts an order r via Quantum Phase Estimation (QPE), and the quantity QPE estimates is an azimuthal phase ϕ on a circle — exactly the coordinate Key Property #2 (Phase Decoupling, Section 3) explicitly isolates away from qg_Z by design. A concrete check confirms this rather

than merely asserting it: simulating a QPE circuit and tracking qg_Z of the eigenstate register throughout shows $qg_Z = -1$ identically, independent of the encoded phase ϕ and of the QPE outcome (companion notebook, Section 16). The information QPE extracts lives entirely in the counting register's measurement distribution, which approximates a point on a circle, not a population bias on an axis.

Scope boundary: This is stated as a scope boundary rather than a future-work item. A unit for phase (a " qg_phi ") is not a straightforward extension of qg_Z — it would need its own inversion relation, its own anchor points, and its own validation from scratch, exactly as qg_Z and qg_S required in Sections 2.1 and 2.2, and is left as an explicitly open question rather than a quick addition. The same boundary applies to quantum counting's phase-estimation step, for the same reason. The honest scope of the qang unit, restated: population-bias problems — Grover's search (Section 4.3), T1/T2 characterization and randomized benchmarking (Section 4.4), and VQE/QAOA (Section 4.1) — are a natural fit; phase-estimation problems are not, by the same design choice that makes qg_Z simple everywhere else it does apply.

Compilation/calibration layer, distinct from the scope boundary above: The scope boundary above is about information, not about gates, and the two should not be conflated. Separately from what qg_Z can read out, the gates used to physically assemble a Shor or quantum-counting circuit — the Hadamards, the QFT's controlled-phase rotations, and any single-qubit gate a controlled-U network decomposes to — are all instances of the $U3_qang$ gate defined in Section 2.4, and the milliqang error-tolerance machinery of Section 4.2 applies directly to characterizing their physical fidelity, independent of which algorithm they appear in. This is a compilation- and calibration-layer connection, not an information-layer one, and it does not revise the finding above: building Shor's circuit out of qang-native gates does not make qg_Z able to read the phase those gates encode.

5. Summary

The qang establishes a pragmatic bridge between geometric qubit parameterization and experimental readout, expressed as a named, normalized unit built on already-established quantities ($\langle\sigma_z\rangle$ and binary Shannon entropy) rather than as a new physical observable.

6. Future Research Directions

The formalization above leaves open several concrete lines of work, organized around the four areas identified as most immediately actionable for the qang framework:

- Quantum-code treatment of the qang as a unit: implement qg as a native unit/type in quantum-computing SDKs (Qiskit, Cirq, PennyLane wrappers), so that circuit parameters, measurement results, and optimizer state can be constructed, converted, and displayed natively in qg / m-qg rather than requiring manual trigonometric conversion at each call site. Status: substantially addressed. The RQangGate and FullRQangGate classes (quang.qiskit_gate, quang.cirq_gate) already provide this for the two-parameter case, and Section 2.4's U3_qang gate completes the family with a third free parameter (λ), verified to machine precision against Qiskit's own U3 definition.
- Formal and empirical validation of the qang: prove the stated bounds, the domain over which qg_Z and qg_S are each invertible (Section 2.2), and error-propagation bounds when converting between θ -space and qg-space; complement this with empirical benchmarking of qang-parameterized optimizers on NISQ hardware and simulators. Status: partially addressed. The invertibility domains (Section 2.2) and the shot-noise error-propagation bound (quang.statistics, $\text{Var}(\text{qg_Z}) = \sin^2\theta/N$) are formalized and validated against simulation. What remains genuinely open is the empirical half of this item: none of the results in this paper (Sections 4.1-4.5) have yet been benchmarked against measurement shots from physical NISQ hardware, only against exact and shot-noise-simulated statevector computation. This is the single most important remaining gap before treating any qg-space result as hardware-validated rather than simulation-validated.
- Characterization of qang gradients: study and, where possible, regularize the coordinate singularity at $\theta = 0, \pi$ identified in Section 4.1, and compare the convergence behavior of qg-space optimization against standard θ -space parameterization and quantum natural gradient methods. Status: addressed. quang.gradients implements both the clipped-inverse-Jacobian and Tikhonov regularizations described here, and a toy VQE benchmark comparing qg-space to θ -space optimization is included in the reference implementation.
- Generalization of qang probabilities: extend the binary framework to POVMs and multi-outcome measurements, to mixed states (where, unlike the pure-state case treated here, the von Neumann entropy of ρ is no longer trivially zero), and to multi-qubit systems via tensor-product projection profiles. Status: addressed. quang.mixed covers POVMs and mixed-state qg_Z via $\text{Tr}(\rho\sigma_z)$, and quang.multiqubit covers tensor-product marginal profiles and joint qg_S, including its use as an operational entanglement witness. Section 2.4 extends the same mechanism to qudit systems via $\text{su}(d)$ Cartan-generator factorization.

6.1 Generalizing the Phase-Estimation Scope Boundary

Section 4.5 established, via direct circuit simulation, that qg_Z of the eigenstate register in a Quantum Phase Estimation (QPE) circuit is identically -1 regardless of the encoded phase, and used this to explain why Shor's algorithm does not admit a qg reformulation. That argument is not specific to Shor: it depends only on QPE's internal structure (an azimuthal phase read out via a counting register), not on what the phase happens to encode. The same degeneracy therefore applies to every algorithm built on QPE as a subroutine, including the HHL algorithm for linear systems (Harrow, Hassidim, & Lloyd, 2009), quantum counting (Brassard, Høyer, & Tapp, 1998), and phase-estimation-based quantum chemistry (Mcardle et al., 2020, already cited in Section 6 above for a different reason). Stated once at this level of generality, the scope boundary of Section 4.5 is a property of the phase-estimation primitive itself, not a list of unrelated negative results to be re-derived algorithm by algorithm.

6.2 Candidate Algorithms: Likely Population-Bias Fits

Two families of algorithms share Grover's defining structure -- a final measurement in the computational basis that resolves a population bias, not a phase -- and have not yet been checked against the qg framework. The Deutsch-Jozsa and Bernstein-Vazirani algorithms (Deutsch & Jozsa, 1992; Bernstein & Vazirani, 1997) both terminate in exactly this kind of measurement (determining "constant vs. balanced", or recovering a secret bitstring), and are a natural, low-cost extension of Section 4.3's treatment: verifying them would broaden "algorithms qg reformulates" beyond a single example. A second, higher-value candidate is stabilizer-code syndrome extraction in quantum error correction (Gottesman, 1997): syndrome measurements are ± 1 -valued parity checks, formally identical in structure to the qg_Z -valued measurements already treated in Section 4.4 for T1/randomized benchmarking. This would be the first connection between the qg unit and quantum error correction, an area the paper does not currently address at all, and is a stronger candidate for future work than the two-algorithm check above.

6.3 Candidate Algorithms: Boundary Cases

Amplitude Estimation (Brassard, Høyer, Mosca, & Tapp, 2002) generalizes Grover's rotation geometry -- the same angle θ_0 from Section 4.3 -- but reads out its result via QPE rather than a direct measurement. Its rotation stage is therefore qg_Z -native by the same argument as Section 4.3, while its readout stage falls under the Section 6.1 scope boundary; it should be

documented as a genuine mixed case rather than assigned wholesale to either category. Quantum walks (discrete- and continuous-time) are a second boundary case: some multi-outcome walk distributions may be expressible through the multi-outcome generalization already present in `qang.mixed`, but this has not been checked against a specific walk construction and is recorded here as an open question rather than a claim.

Deferred: A closed-form phase unit $\text{qg_Phi}(\phi) := e^{(i*2*\pi*\phi)}$, built directly from Euler's formula and invertible over its full domain (unlike `qg_Z` or `qg_S`), was explored alongside this revision's Section 2.4 work and is documented in the companion RFC addendum. It is deliberately not promoted to a numbered section of this paper: as noted in Section 4.5, a phase unit needs its own anchor points and validation from scratch, and is left as explicitly deferred future work rather than folded in prematurely.

Code Availability

A reference implementation of the `qang` unit (`qg_Z`, `qg_S`, their inversion relations, and the milliqang subunit), native Qiskit and Cirq gates, closed-form transformations (including the Grover-diffuser and transition-probability helpers used in Section 4.3), and a Bayesian estimator are available at <https://github.com/Viny2030/qang>, together with a test suite that reproduces the anchor values in Tables 1 and 2, the gradient singularity discussed in Section 4.1, and the results in Sections 4.3-4.5. A companion executed notebook (`qang_full_reference.ipynb`, in the repository's `notebooks/` folder) walks through every result in this paper end to end, including Sections 4.3-4.5 (notebook Sections 14-16). A companion RFC proposes upstreaming a native `qang` module into Qiskit and Cirq.

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